Using Lookup Metrics

Availability of Lookup Metrics

The `Lookup By` operator is only available when using the Railgun engine, and when explicitly enabled via configuration. To activate this functionality, reach out to the Feedzai Customer Success (CS) team.

Lookup By metrics provide a powerful extension to standard Group By metrics, by allowing users to choose which group's aggregation to retrieve for an event, based on fields other than the groupBy fields.

This page describes how to create a Lookup By metric in Pulse, and provides a practical example of how to combine it with standard Group By metrics to calculate new derived fields.

Creating Lookup By Metricsâ

To create a Lookup By metric in Pulse, follow these steps:

1. In the **Data Science** tab, open the features plan.
2. Drag a Metrics operator to the plan if there is none yet.

3. Edit the Metrics operator:
4. Select the [Add new Metric] button to add a new metric to the Metrics operator.
5. Select the **Group By** fields over which to calculate aggregated metrics. For example, *receiver_ID*:
6. Select the [Add look-up by] button to activate the **Lookup By** input field.
7. Select the appropriate look-up fields from the available options. For example, *sender_ID*:

   warning

   The lookup fields must be in the same number and of the same data type as the group by fields, and arranged in the same order.

8. Define the remaining sections (Window, Aggregation, Description) as needed, according to the regular process for [Creating Metric](#).
9. Save the Metrics operator with look-up configuration.
10. Save the Plan with new metrics.

Combining Lookup By with Group By Metricsâ

Lookup By metrics are often useful when combined with standard Group By metrics. To do so, add both types of metrics to the same Metrics operator, and combine them with a [Map operator](#).

The following example demonstrates how to set up such a combination of metrics for the use case of calculating an accountâ□□s recent running balance at each transaction. It assumes that the input data stream contains at least the fields `sender_ID`, `receiver_ID`, and `amount` (in addition to the obligatory timestamp).

1. Create (if necessary) a **Metrics** operator as described above, and edit it.
2. Select the [Add new Metric] button and add the first metric to the Metrics operator. This will be a regular **Group By** metric, to get the total amount sent by the transaction's sender.

    1. In the **Group By** section, select the *sender_ID* field.
    2. In the **Window** section, set the window to *7 days*.
    3. In the **Aggregation** section, select the *sum* function over the *amount* field.
    4. Set the **Name** of the metric to *total_sent_1w*.

3. Select the [Add new Metric] button and add the second metric to the Metrics operator. This will be a **Lookup By** metric, to get the total amount received by the transaction's sender.

    1. In the **Group By** section, select the *receiver_ID* field; activate the Lookup By input area as described above, and add the *sender_ID* field to it.
    2. In the **Window** section, set the window to *7 days*.
    3. In the **Aggregation** section, select the *sum* function over the *amount* field.
    4. Set the **Name** of the metric to *total_received_1w*.

4. Review the **Metrics** operator with the two metrics. Confirm that the grouping and lookup fields are correctly set for each metric by hovering over the ð□□□paperclip icon next to each field in the **Group By** column.

    Then, [Save] the Metrics operator.

5. Add a **Map** operator to the plan by dragging the [Map] button to the plan's diagram, after the Metrics operator.
6. Edit the Map operator. In the **Fields To Add** tab, create a new field as follows:

    1. Set the **Output Field Name** to *balance_1w*,
    2. Set the **Expression** to `total_received_1w` - `total_sent_1w`.
    3. Leave the **Type** as the default *Numeric*.

    Finally, [Save] the Map operator.

By completing these steps, the plan will now add the *balance_1w* field to each event, representing the account's running balance over the past week at the time of the transaction.